

Bayes Optimal Classifier & Gibbs Algorithm — Lecture Notes

Part 1. Motivation

- **MAP (Maximum A Posteriori):** chooses the most probable hypothesis.
 - **Limitation:** MAP only uses one hypothesis, ignoring all others.
 - **Bayes Optimal:** considers all hypotheses, weighted by their posterior probabilities.
 - **Goal:** make the most probable classification of a new instance, not just select the most probable hypothesis.
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Part 2. MAP vs Bayes Optimal

- **MAP rule:**

$$h_{\text{MAP}} = \arg \max_{h \in H} P(h|D)$$

Predict using this single hypothesis.

- **Bayes Optimal rule:**

$$v^* = \arg \max_{v_j \in V} P(v_j|D)$$

where

$$P(v_j|D) = \sum_{h \in H} P(v_j|h)P(h|D).$$

👉 Bayes Optimal = weighted vote of all hypotheses.

Part 3. Worked Example (Step-by-Step)

Hypotheses and Posteriors

$$P(h_1|D) = 0.4, \quad P(h_2|D) = 0.3, \quad P(h_3|D) = 0.3$$

Predictions for new instance x : - $h_1(x) = \oplus$ (Positive) - $h_2(x) = \ominus$ (Negative) - $h_3(x) = \ominus$ (Negative)

MAP Classification

- Choose h_1 because it has highest posterior (0.4).
- MAP prediction: **Positive** ($\oplus$).

Bayes Optimal Classification

We compute class probabilities by weighting each hypothesis prediction by its posterior:

- For Positive ($\oplus$):

$$P(\oplus|D) = P(\oplus|h_1)P(h_1|D) + P(\oplus|h_2)P(h_2|D) + P(\oplus|h_3)P(h_3|D) = 1(0.4) + 0(0.3) + 0(0.3) = 0.4$$

- For Negative ($\ominus$):

$$P(\ominus|D) = P(\ominus|h_1)P(h_1|D) + P(\ominus|h_2)P(h_2|D) + P(\ominus|h_3)P(h_3|D) = 0(0.4) + 1(0.3) + 1(0.3) = 0.6$$

Bayes Optimal chooses **Negative** ($\ominus$).

Part 4. Properties of Bayes Optimal

- Maximizes probability of correct classification.
 - No other classifier using the same hypothesis space can do better on average.
 - Predictions may **not correspond to any single hypothesis** in H . It effectively acts like a new "meta-hypothesis".
 - Viewed as working in an expanded space H' that allows combinations of hypotheses.
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Part 5. How to Pick Randomly from a Distribution

When we say "pick a hypothesis at random from the posterior distribution," we mean:

We have probabilities:

$$P(h_1|D) = p_1, \quad P(h_2|D) = p_2, \quad \dots, \quad P(h_k|D) = p_k$$

Inverse Transform Sampling

1. Generate a uniform random number $u \sim U[0, 1]$.
2. Compute cumulative distribution:

$$F(j) = \sum_{i=1}^j p_i$$

3. Choose h_j if

$$F(j-1) < u \leq F(j).$$

Example

- Posteriors: $p_1 = 0.4, p_2 = 0.3, p_3 = 0.3$.
- CDF: $F(1) = 0.4, F(2) = 0.7, F(3) = 1.0$.
- If $u = 0.25 \rightarrow$ choose h_1 .
- If $u = 0.55 \rightarrow$ choose h_2 .
- If $u = 0.85 \rightarrow$ choose h_3 .

👉 This is the math behind “picking randomly according to a distribution.”

Part 6. Gibbs Algorithm

- **Problem:** Bayes Optimal is computationally expensive.
 - **Gibbs Algorithm (Oppor & Haussler, 1991):**
 - Pick one hypothesis h at random according to posterior distribution.
 - Predict using $h(x)$.
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Part 7. Performance of Gibbs Algorithm

- **Expected error of Gibbs:**

$$\mathbb{E}_{h \sim P(h|D)}[\text{err}(h, x)] = \sum_{i=1}^k P(h_i|D) \cdot \text{err}(h_i, x)$$

- **Guarantee (Haussler et al., 1994):**

$$\mathbb{E}[\text{Error}_{\text{Gibbs}}] \leq 2 \cdot \mathbb{E}[\text{Error}_{\text{BayesOptimal}}]$$

- Assumptions: target concepts drawn from prior distribution.
 - With uniform prior: random choice from version space has $\leq 2 \times$ Bayes Optimal error.
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Part 8. Intuition & Takeaway

- **MAP:** best single hypothesis.
- **Bayes Optimal:** weighted vote of all hypotheses (gold standard).
- **Gibbs:** pick one hypothesis at random (cheap approximation).
- **Guarantee:** Gibbs $\leq 2 \times$ Bayes Optimal error.

👉 **Student takeaway:** Bayes Optimal is the best, but Gibbs shows that even a simple randomized method can stay close to optimal.